روش‌های پیشرفته نمونه برداری در جنگل

نمونه برداری خطی
Line sampling

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مدل مفهومي نمونه برداري

- Protocol

طراحي نمونه برداري
Sampling design

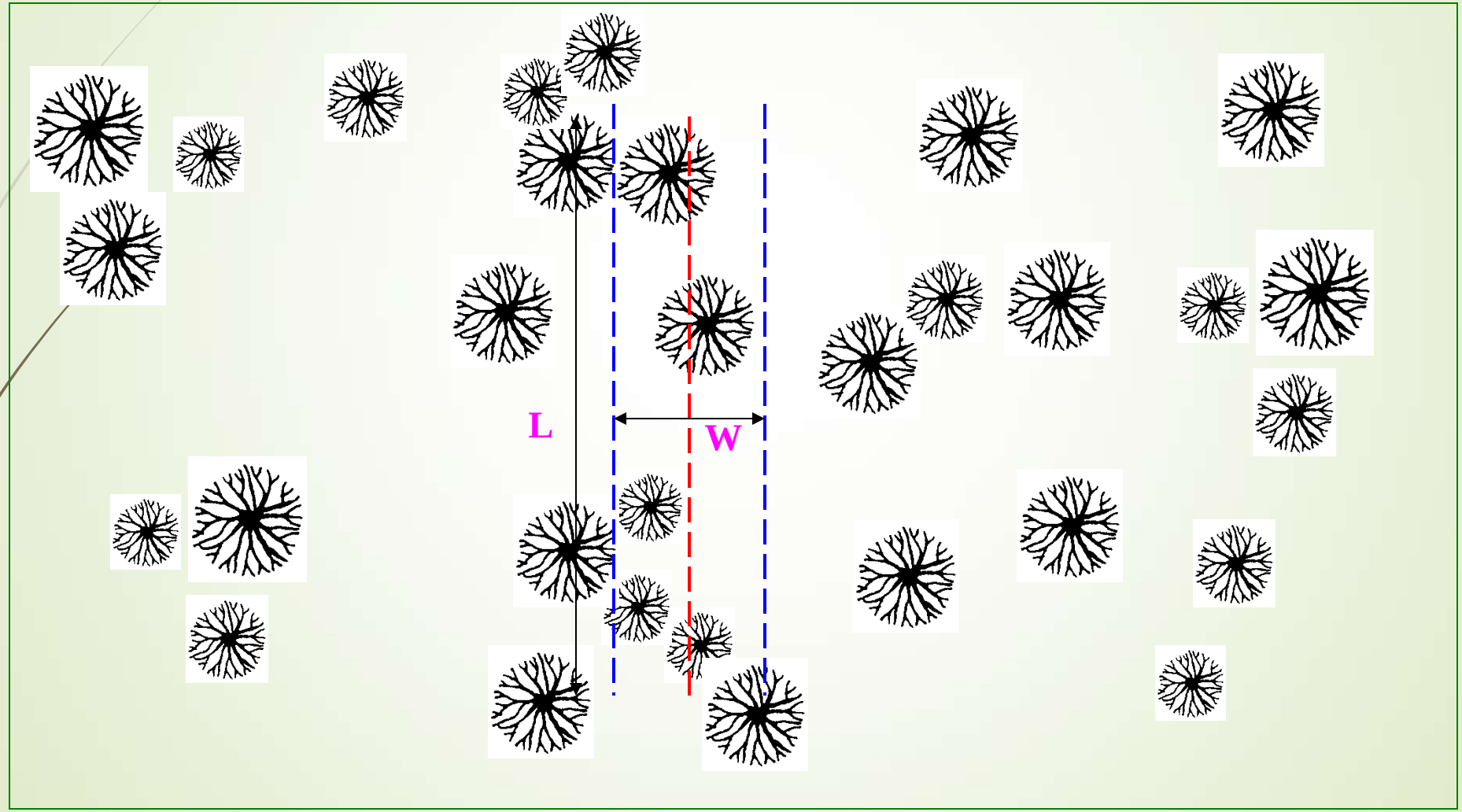
- روش نمونه برداري
  Sampling Method
  - نمونه برداري نقطه‌اي Point Sampling
  - نمونه برداري فاصله‌اي Distance Sampling
  - نمونه برداري خطي Line Sampling
  - نمونه برداري قطعه‌اي Plot Sampling
- الگوي نمونه برداري
  Sampling patterns
  - تصادفي ساده Simple Random S.
  - بلوك تصادفي Block Random S.
  - سيستماتيک تصادفی ساده Systematic S.
  - سيستماتيک مبتنی بر خطوط ميزان (Contour sampling)
- اجتماع نمونه
  Sampling clusering
  - نمونه واحد Single s.
  - نمونه خوشه‌اي Cluster S.
- استراتژي نمونه برداري
  Sampling strategies
  - مونه بندي
    - پيشا مونه بندي(Prestratification)
    - پسا مونه بندي (Poststratification)
  - نمونه برداري چند گانه
    Multiphase S.
    - Two phase sampling with regression estimator
    - Two phase sampling with ratio estimator
    - Two phase sampling for stratification
    - .....
  - نمونه برداري چند گامي
    Multistage S.
    - Two stage sampling
    - Three stage sampling
    - .....

روش‌های نمونه برداری خطی

Line sampling methods

- **transect**
  - Line Transect Sampling
  - Horizontal Line Sampling
  - Modified Horizontal Line Sampling
  - Variable Area Transect
  - Modified Variable Area Transect
- **intersect**
  - Line Intercept Sampling
  - K-tree Line Intersect Sampling
  - Fixed Length Line Intersect Sampling (I)
  - Fixed Length Line Intersect Sampling (II)

Line Transect Sampling (based on Fixed Width)



LTS Estimator

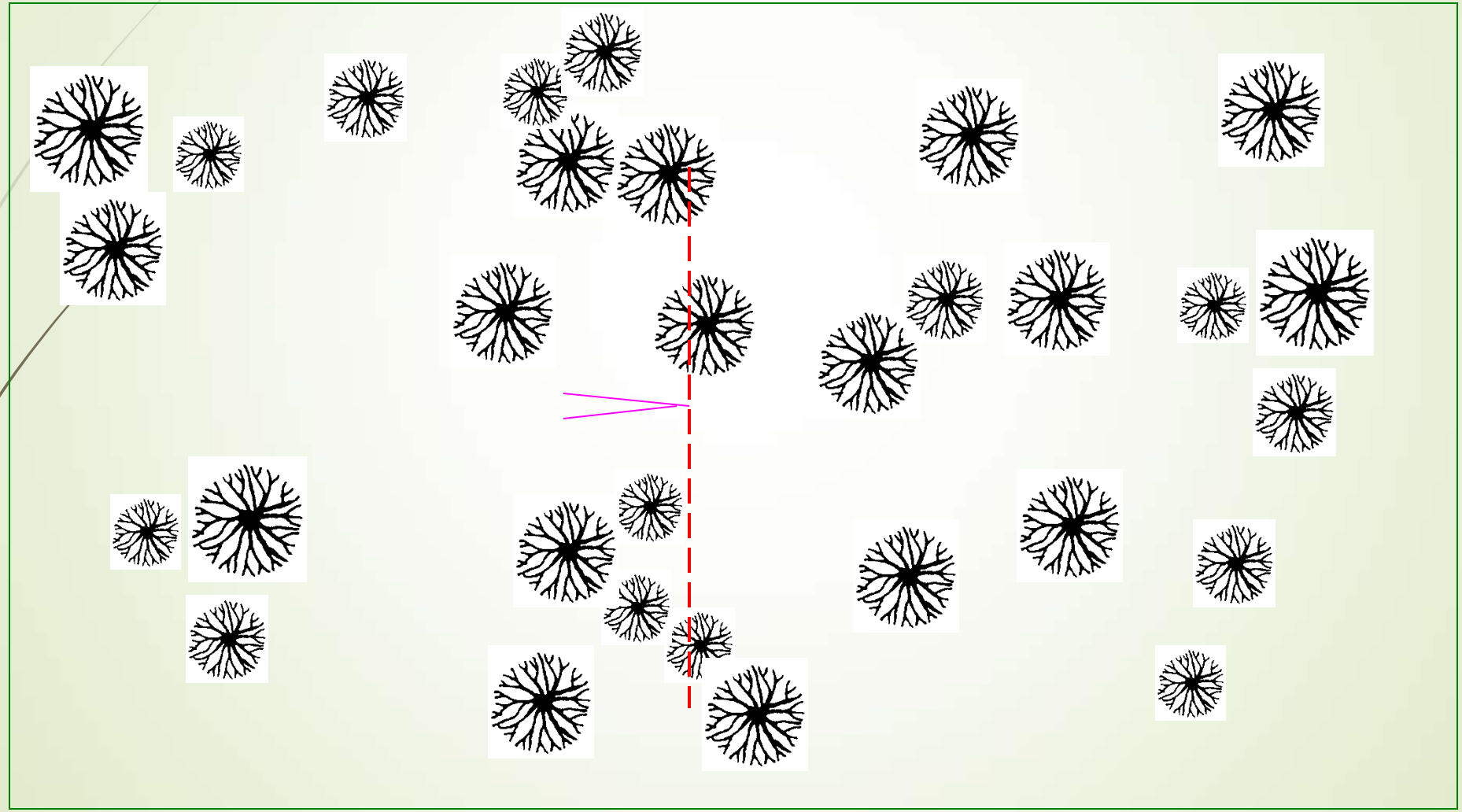
$$\hat{y}_i = ef \times \frac{\sum_{j=1}^{k} y_{ij}}{a_i}$$

$$a_i = W_i \times L_i$$

$$\hat{\bar{y}} = \frac{\sum_{i=1}^{n} \hat{y}_i}{n}$$

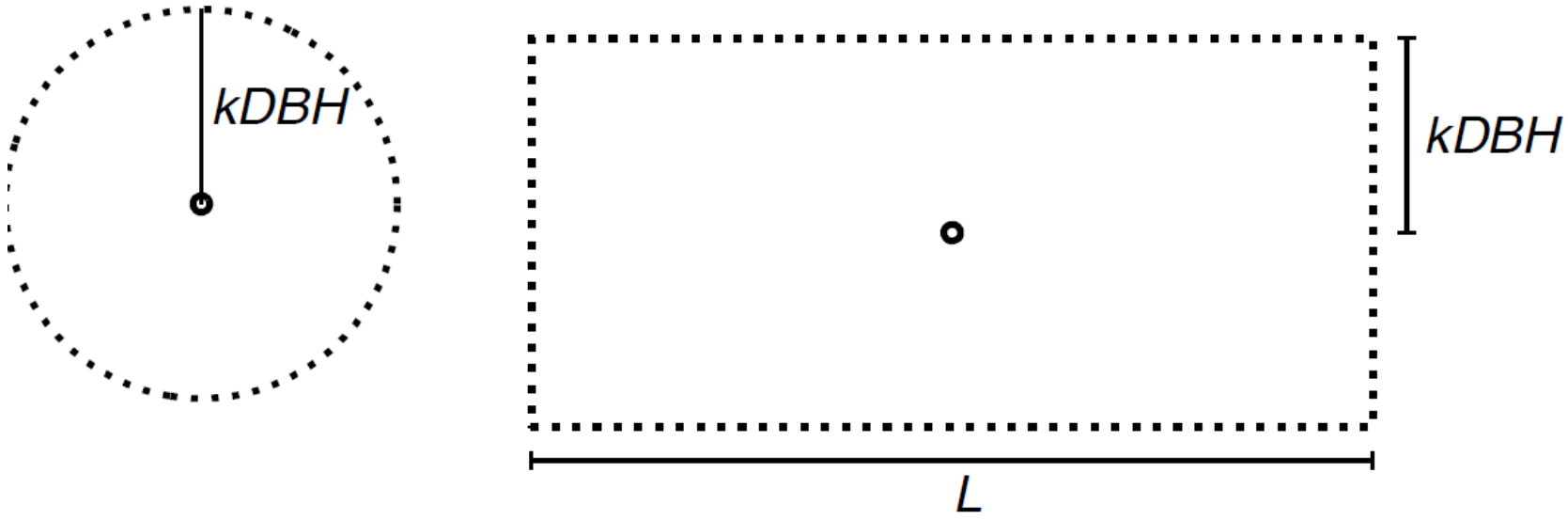
$$S_y^2 = \frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n-1}$$

Horizontal Line Sampling (HLS)



Horizontal Line Sampling (HLS)

Example inclusion zones for horizontal point sampling (HPS) and horizontal line sampling (HLS). The parameter *k* is the characteristic factor of the angle gauge; the limiting distance for a tree is *k* × DBH. *L* represents the length of the sample line in HLS. The orientation of the HLS inclusion zone is determined by the orientation of the sample line.



HLS estimator

$$\hat{N} = Ac\sum_i \frac{1}{a_i}$$

Where

a_i is the area of the inclusion zone for the *i*th tree, and the summation is taken over the tallied trees.

The constant c represents the conversion from the units of a_i to those of A (i.e., 10000 m2/ha).

The quantity c/a_i is the expansion factor for the *i*th tree.

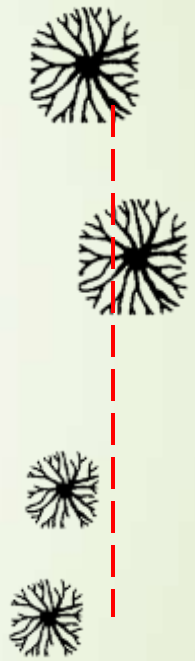
Likewise, the estimator for any total quantity, Y, to which an individual tree contributes y_i units, is

$$\hat{Y} = Ac\sum_i \frac{y_i}{a_i}$$

Tow Problems of HLS

The first problem, from a certain standpoint psychological but hinting at an underlying inefficiency, was the frequent presence of interested trees near the end of an HLS segment but beyond it and not within the “variable width strip”.

The second problem compounded the first: when trees of interest were very close to the end of the HLS segment, it was necessary to determine accurately whether the trees should be included in the sample, using an angle mirror, right angle prism, or an accurate compass.

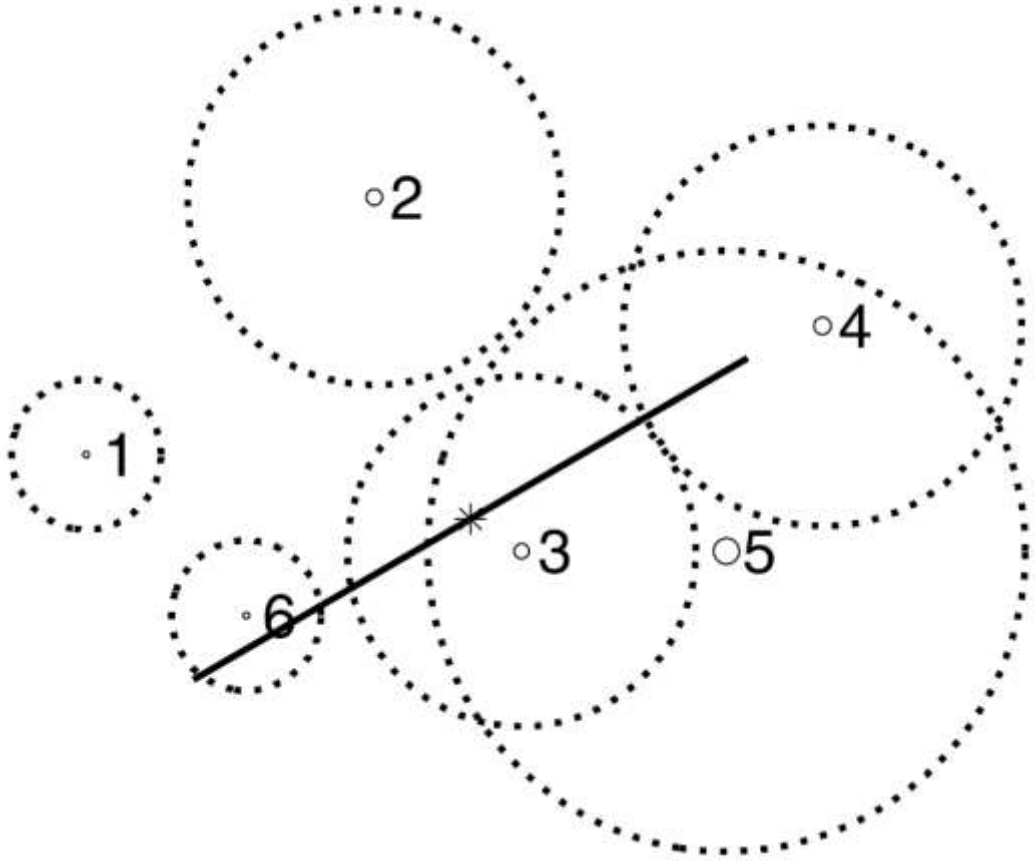


Modified HLS [MHLS]

Implementation of modified horizontal line sampling.

The center of the sample line (*) is located in the field, and the sample line is run at a predetermined distance and bearing. All trees close enough to the line to tally (i.e., those for which the sample line crosses their HPS inclusion zone) are included in the sample. In this example, trees 3, 4, 5, and 6 would be tallied in MHLS.

Ordinary HLS would ignore tree 4, because it is beyond the end of the line. Only trees 3 and 5 would be tallied in HPS.



Modified HLS [MHLS]

Inclusion zone for modified horizontal line sampling. The parameter k is the characteristic factor of the angle gauge; the limiting distance for a tree is $k \times$ DBH. L represents the length of the sample line. The orientation of the sample line determines the orientation of the inclusion zone. If the centre of the sample line falls inside a tree’s inclusion zone, it will be tallied; otherwise, it will not.

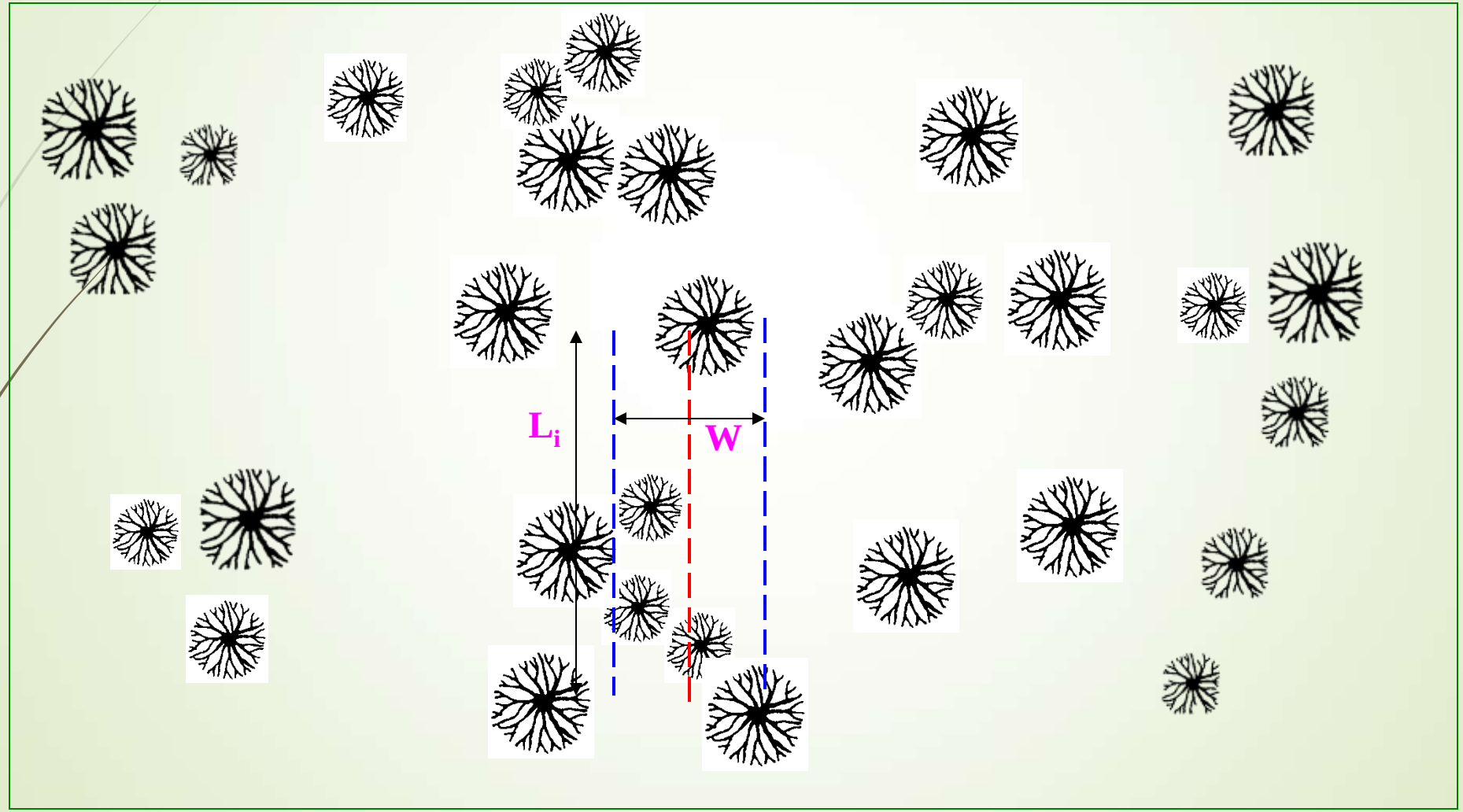
MHLS estimator

Thus, the area of the inclusion zone for the *i*th tree is a quadratic in $k \times$ DBH:

$$a_i = 2kL \times \mathrm{DBH}_i + \pi k^2 \times \mathrm{DBH}_i^2$$

$$\hat{N} = Ac \sum_i \frac{1}{a_i}$$

Variable Area Transect (VAT)



VAT ESTIMATOR

If we let w be the width of the strip transect and L_i be the length searched from the ith random starting point to the rth individual, then the formula for the VAT estimator, which we call the original formula, is:

$$\hat{D}_{\text{orig}} = (nr - 1)/(w \sum L_i)$$

Because this estimator was developed under the assumption that the spatial pattern for the sampled population was random, Parker (1979) also proposed an analog to Morisita's (1957) method for producing an unbiased sample with closest individual sampling in an aggregated spatial pattern. The formula for this Morisita analog is

$$\hat{D}_{\text{mor}} = [(r - 1) \sum (1/L_i)]/(nw)$$

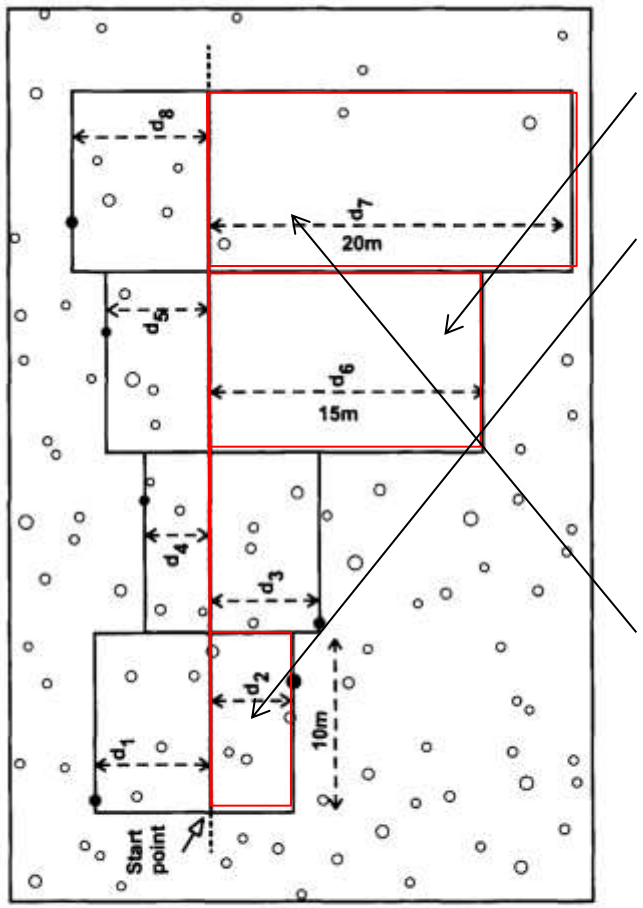
VAT estimator

Parker 1979

$$\widehat{\text{var}}(\hat{D}_{\text{orig}}) = (\hat{D}_{\text{orig}})^2/(nr - 2)$$

$$\widehat{\text{var}}(\hat{D}_{\text{mor}}) = (\hat{D}_{\text{mor}})^2/[n(r - 2)]$$

Modified Variable Area Transect (MVAT)



The sample unit. This is composed of eight variable cells of 10-m-wide that extend from the 40-m transect line. All distances are defined horizontally. Each cell captures five trees, or less, and the distance to the most distant tree included (filled in the figure) is recorded (d_1, d_2,... etc.). The maximum distance to search in each cell before deciding it is "empty" is 15 m (see d_4). The maximum distance to search to collect up to five stems is 20 m (see d_7)

1. If a horizontal distance L_{min} is traveled without encountering any trees, that cell is tallied (i.e. scored) as empty (a zero).
2. If at least one tree is tallied before reaching L_{mjn}, and some maximum number of trees r_{max} is tallied before reaching a maximum horizontal distance L_{max}, the cell is recorded as containing r trees, and its length is recorded as the distance from the centre-line to tree r_{max} (this measure is recorded to the stem centre, not the nearest point).
3. If L_{max} is reached before r_{max} trees have been tallied, then sampling stops. The cell is recorded as containing the number of trees tallied so far, and its length is recorded as L_{max}.

MVAT ESTIMATOR

For a cell, we can distinguish three situations:

1. The transect is run out for a horizontal distance L_{empty} , and no trees are encountered. The cell is tallied as empty. In this case, the total density estimate for the cell, X., equals 0.

2. The maximum number of trees r_{max} is encountered before L_{max} is reached; the total length of transect run in the cell is L_i. The total density estimate corresponds to the variable-area transect estimate for a single cell:

   $$\mathbf{X_i=(r_{max}- 1)/ (wL_i)}$$

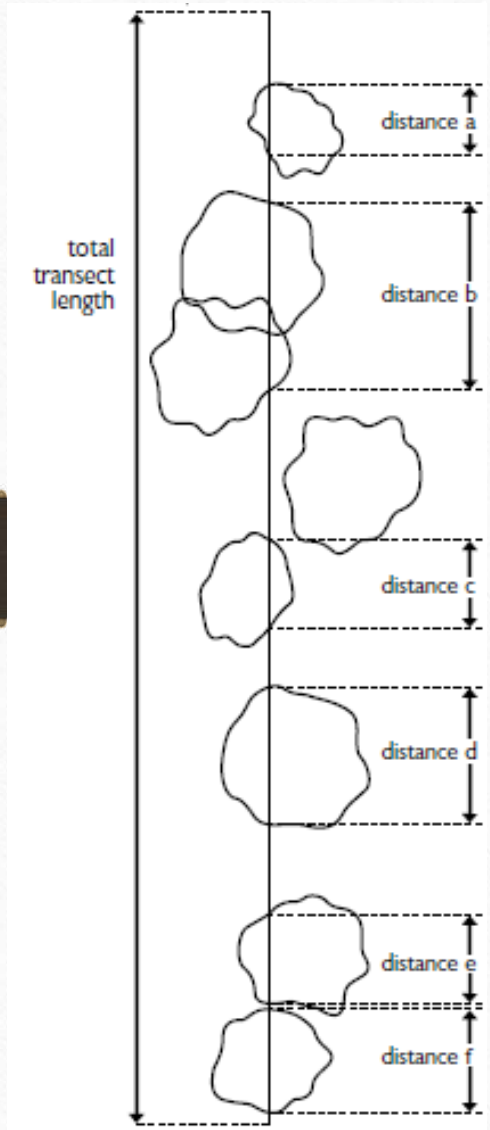
3. The cell is extended to its maximum horizontal length L_{max}, and less than the maximum number (in our implementation, five) trees are tallied. If r_i trees are tallied, the total density estimate is

   $$\mathbf{X_i = r_i / (wL_{max}).}$$

روش‌های نمونه برداری خطی

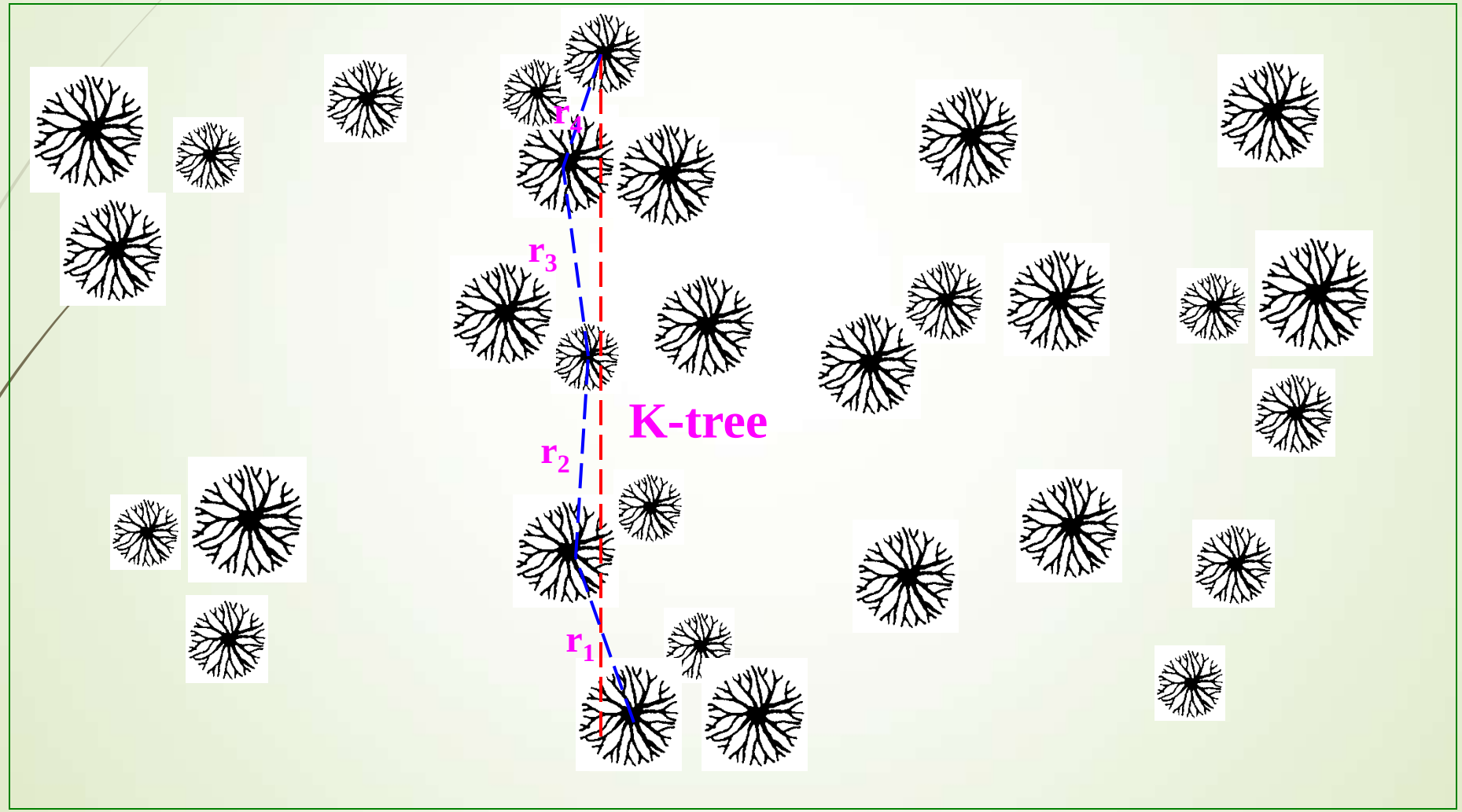
- **intersect**
  - **Line Intercept Sampling**
  - **K-tree Line Intersect Sampling**
  - **Fixed Length Line Intersect Sampling (I)**
  - **Fixed Length Line Intersect Sampling (II) (Strong method 1966)**

Line Intercept Sampling



$$\%\text{cover} = \frac{\text{distance a+b+c+d+e+f}}{\text{total transect length}}$$

K-tree Line Intersect Sampling



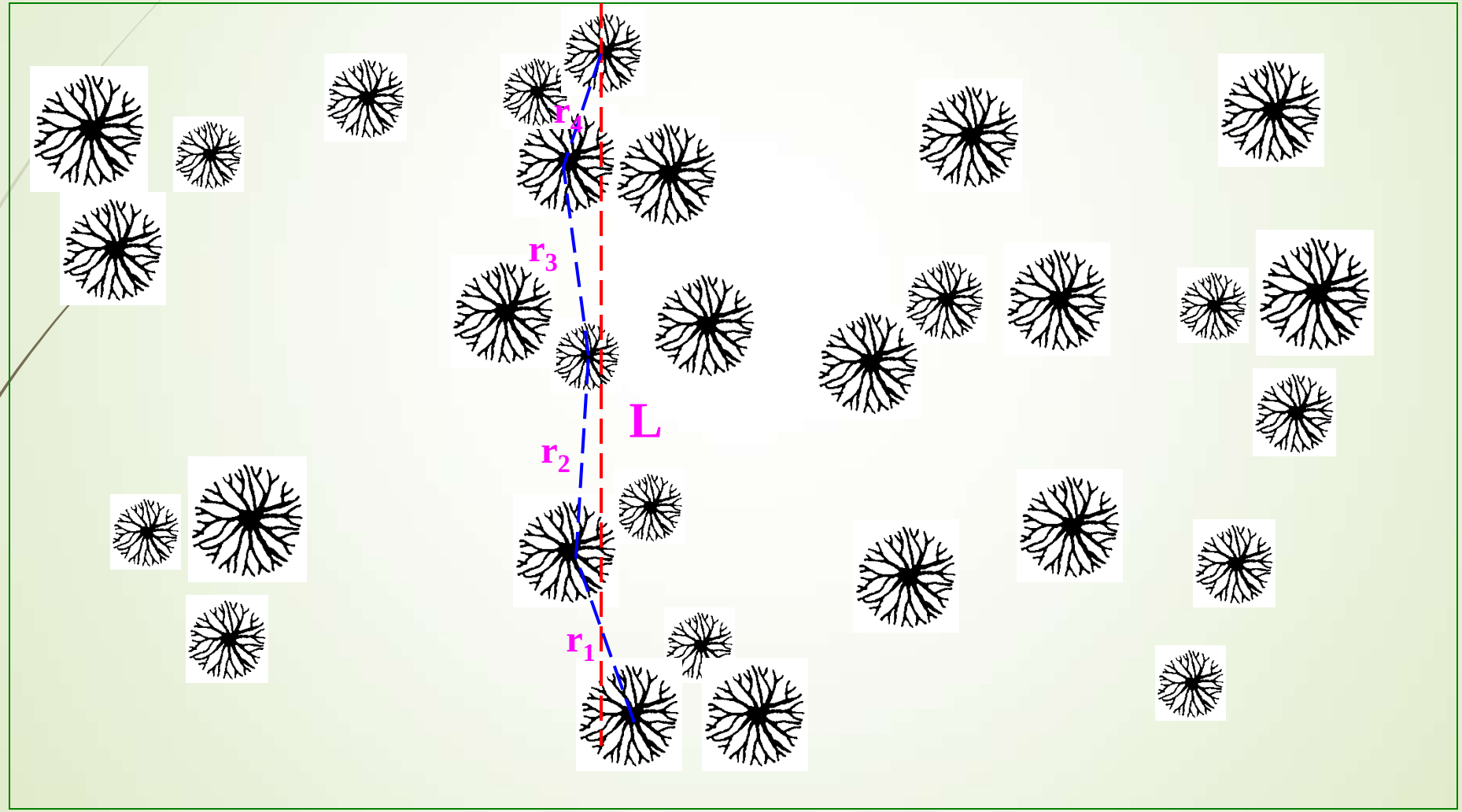
برآورد کننده

$$\bar{r}_i = \frac{\sum_{j=1}^{k} r_j}{k_j}$$

$$\hat{N}_i\,(ha) = \frac{10000}{\bar{r}_i^{\,2}}$$

$$\hat{\bar{N}} = \frac{\sum_{i=1}^{n} \hat{N}_i}{n}$$

Fixed Length Line Intersect Sampling (I)



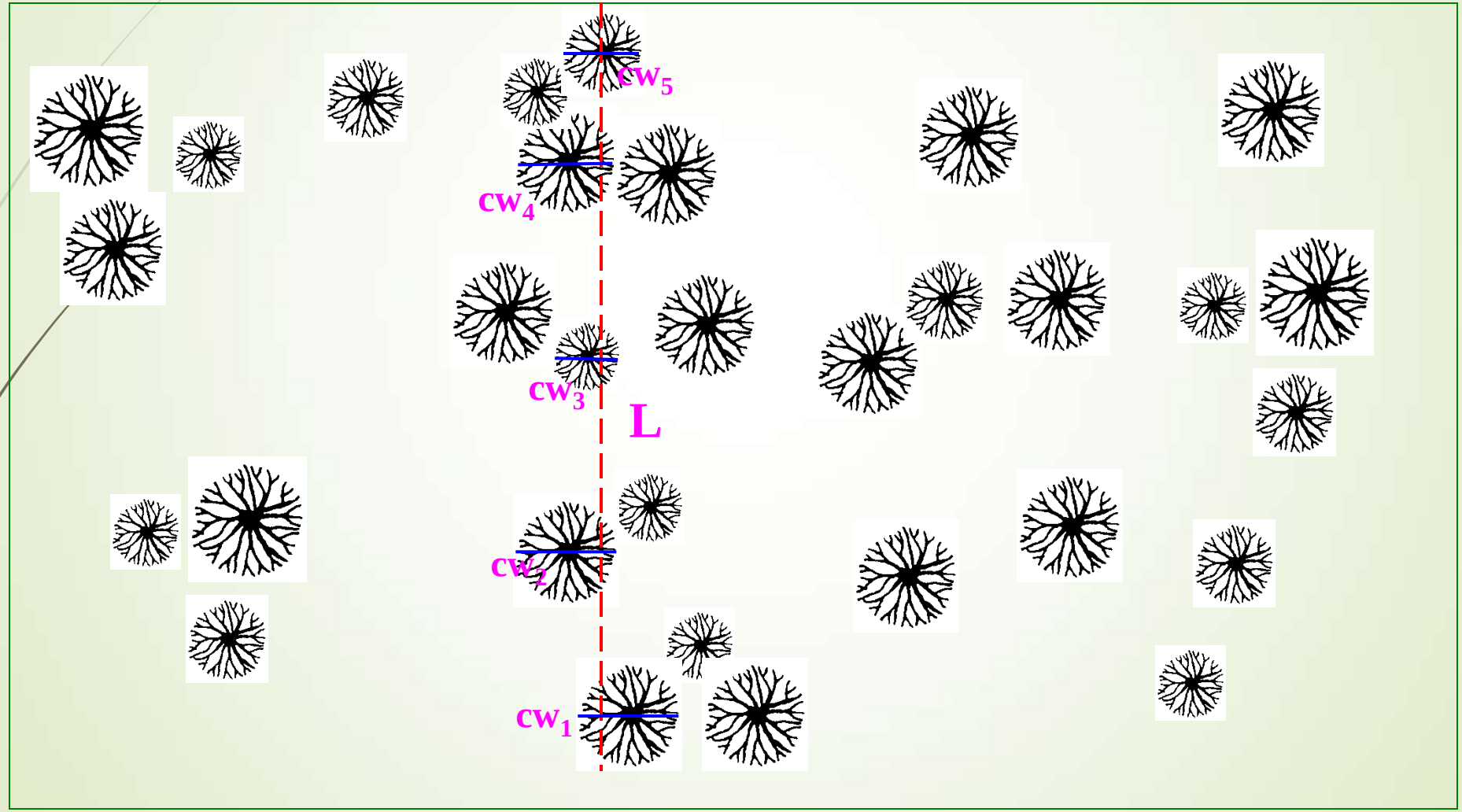
برآورد کننده

$$\bar{r}_i = \frac{\sum_{j=1}^{k} r_j}{\sum_{j=1}^{k} k_j}$$

$$\hat{N}_i\,(ha) = \frac{10000}{\bar{r}_i^{\,2}}$$

$$\hat{\bar{N}} = \frac{\sum_{i=1}^{n} \hat{N}_i}{n}$$

Fixed Length Line Intersect Sampling (II)

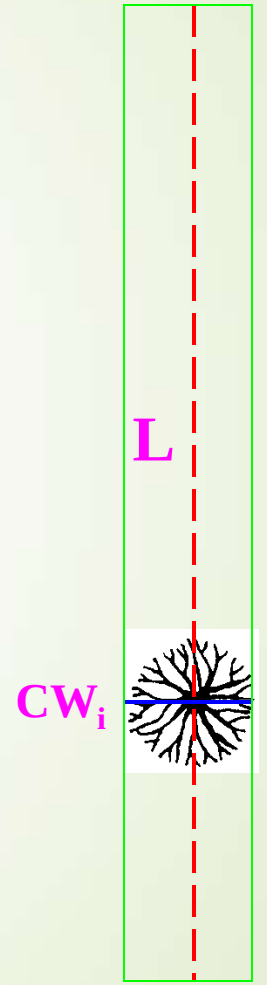


برآورد کننده

$$a_i = L \times CW_i$$

$$\hat{N}_j = 10^4 \times \sum_{i=1}^{n_j} \frac{1}{a_i}$$

$$\hat{Y}_j = 10^4 \times \sum_{i=1}^{n_j} \frac{Y_i}{a_i}$$



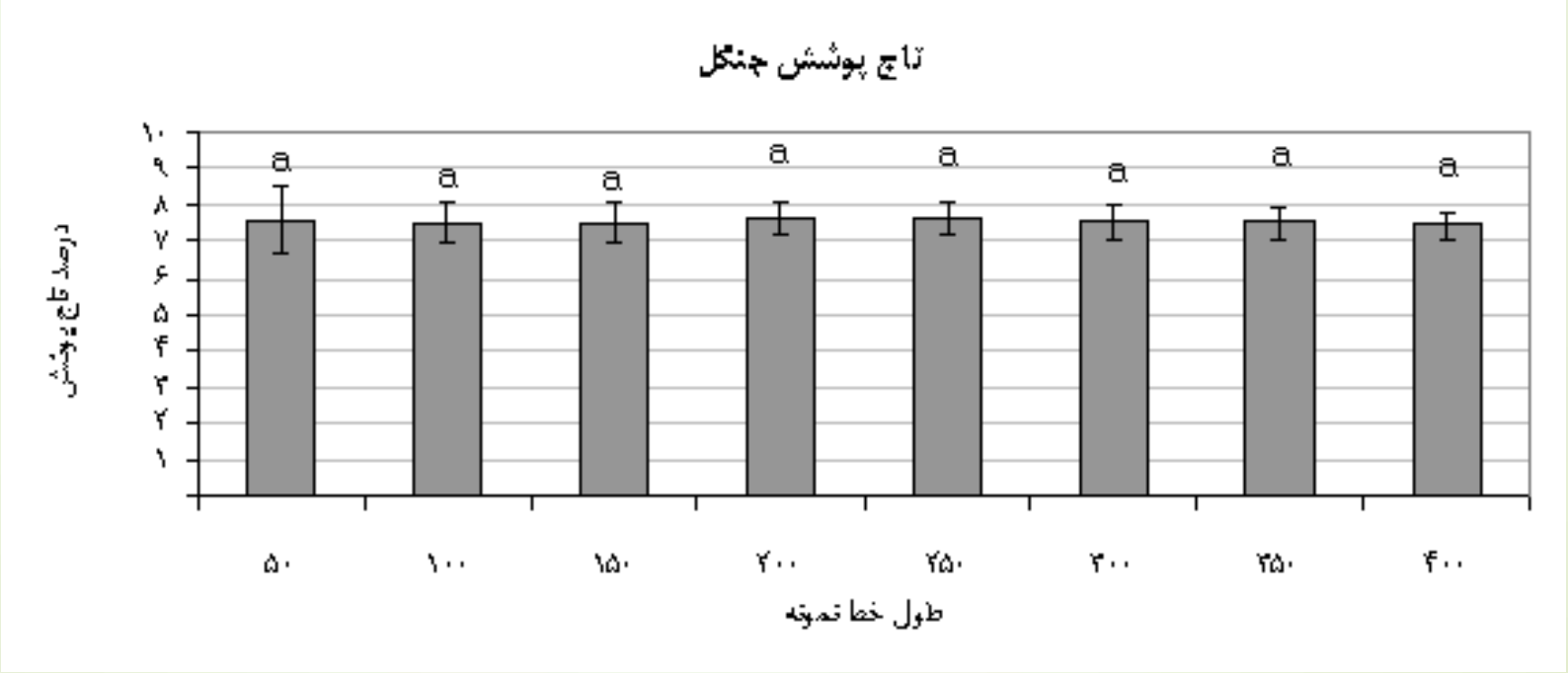
صحت روش خط نمونه در برآورد تاج پوشش و تراکم جنگل‌های زاگرس

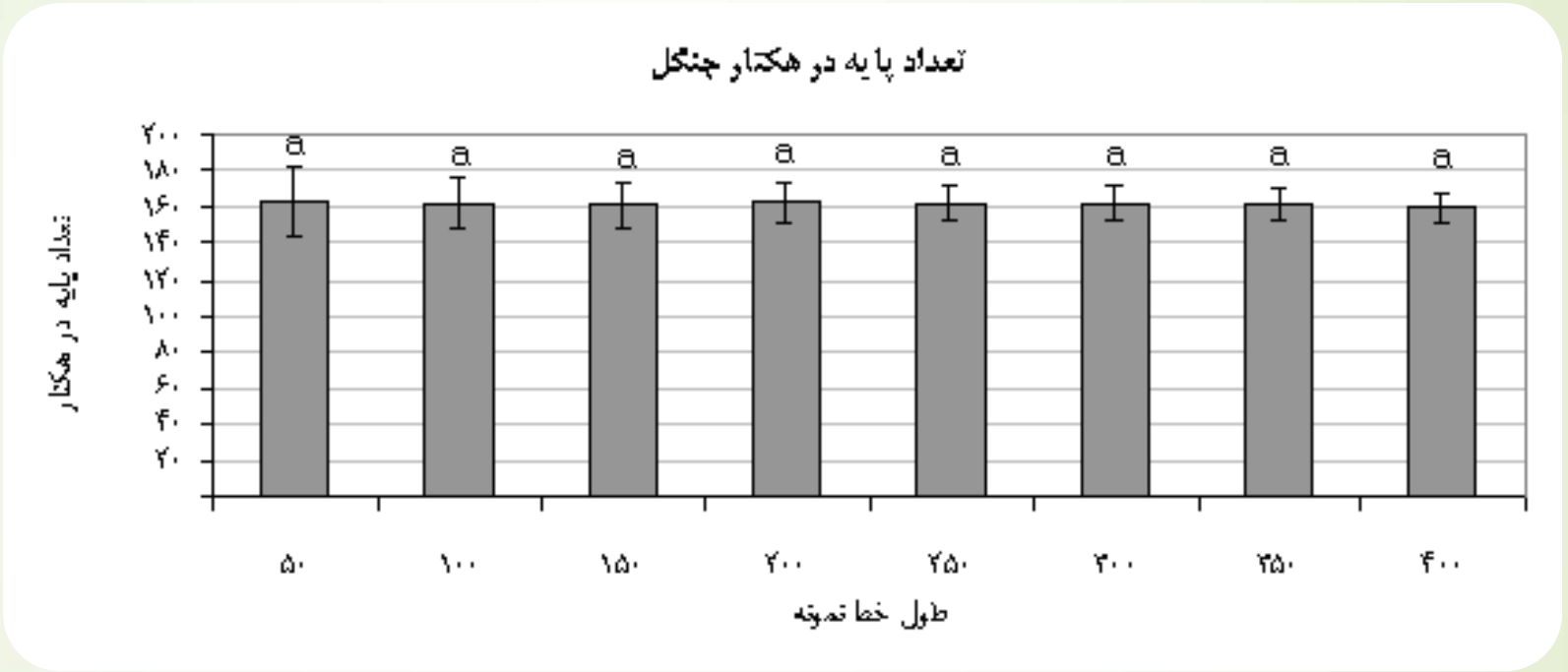
سهرابی و همکاران، 1391

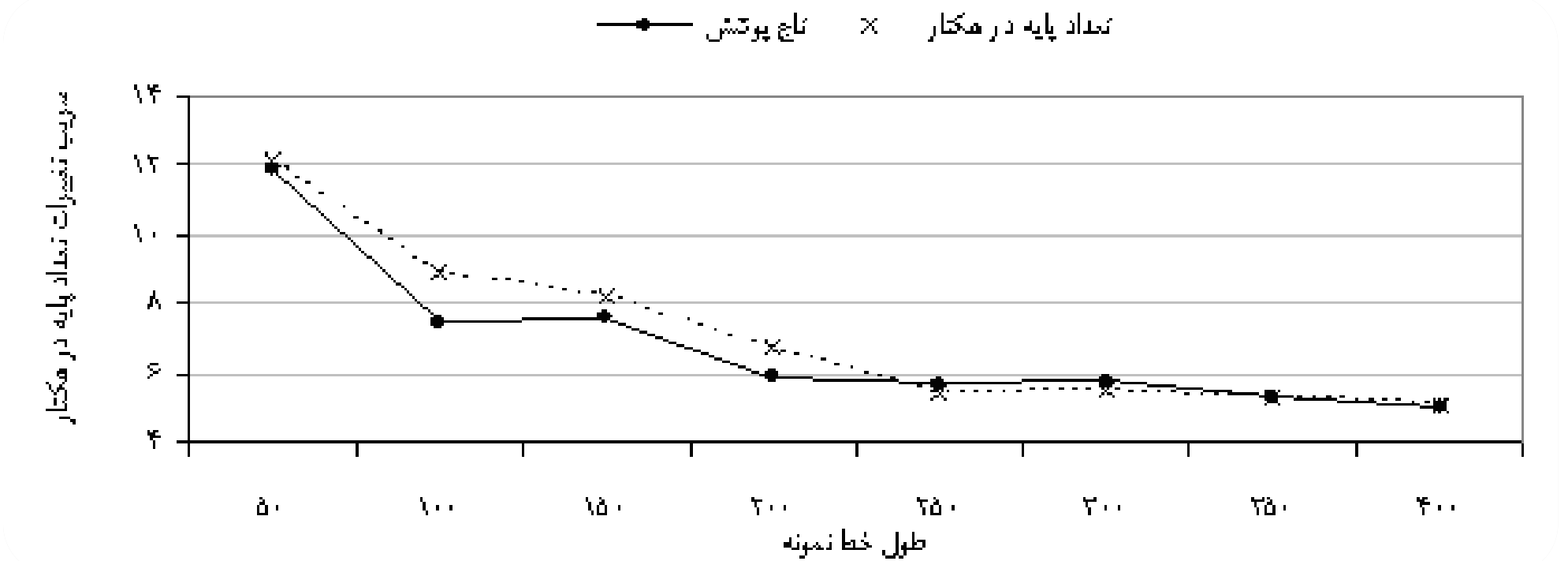
$$Study\ \ Area = 53ha$$

$$100\,sample\ (each: n = 100)$$

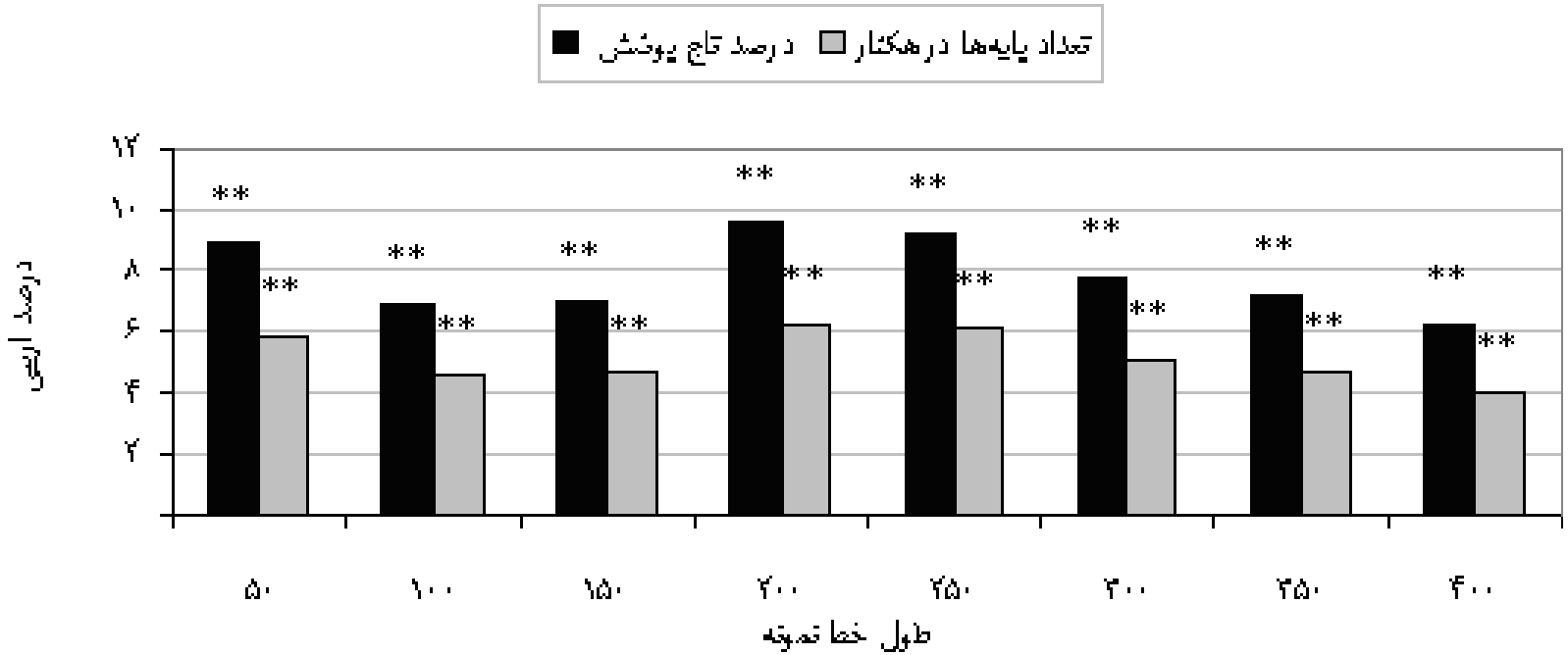
$$L: 50 - 400 (span: 50)$$




تعداد پایه در هکتار جنگل
تعداد پایه در هکتار
۲۰۰
۱۸۰
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طول خط نمونه

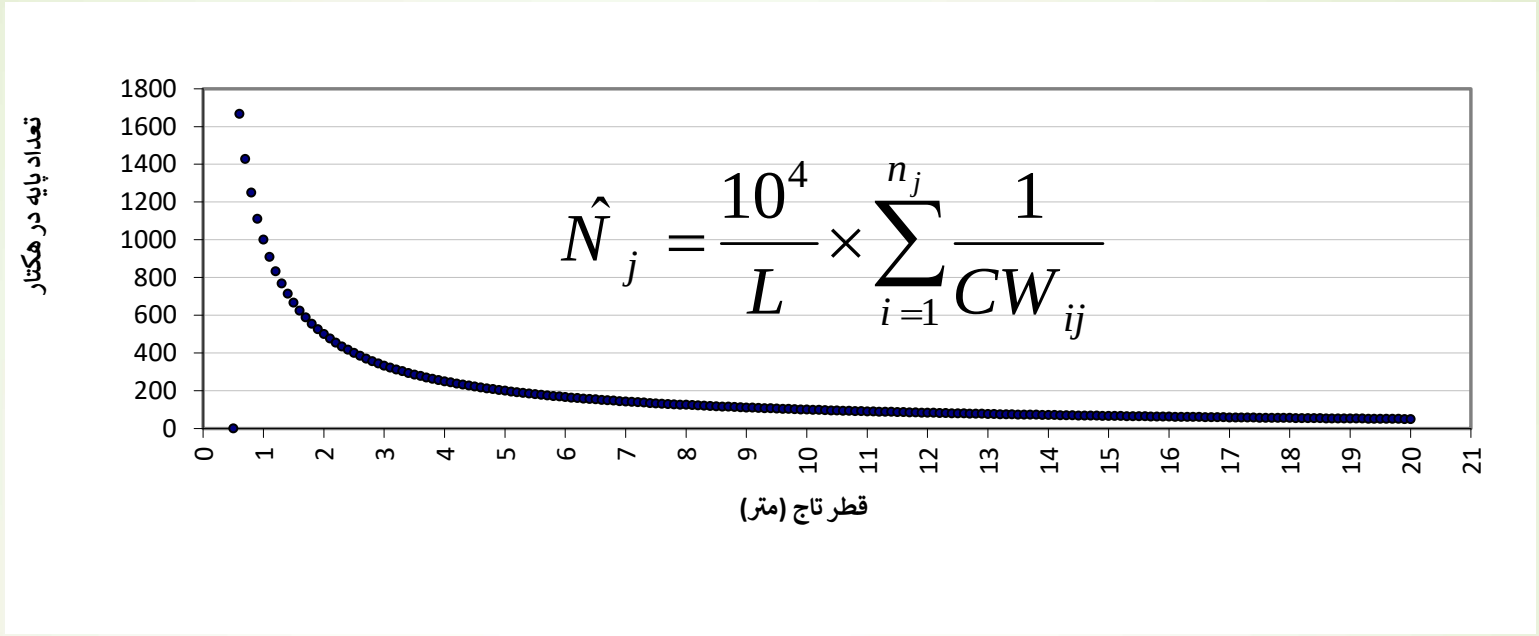

تعداد پایه در هکتار
تاج پوشش
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طول خط نمونه

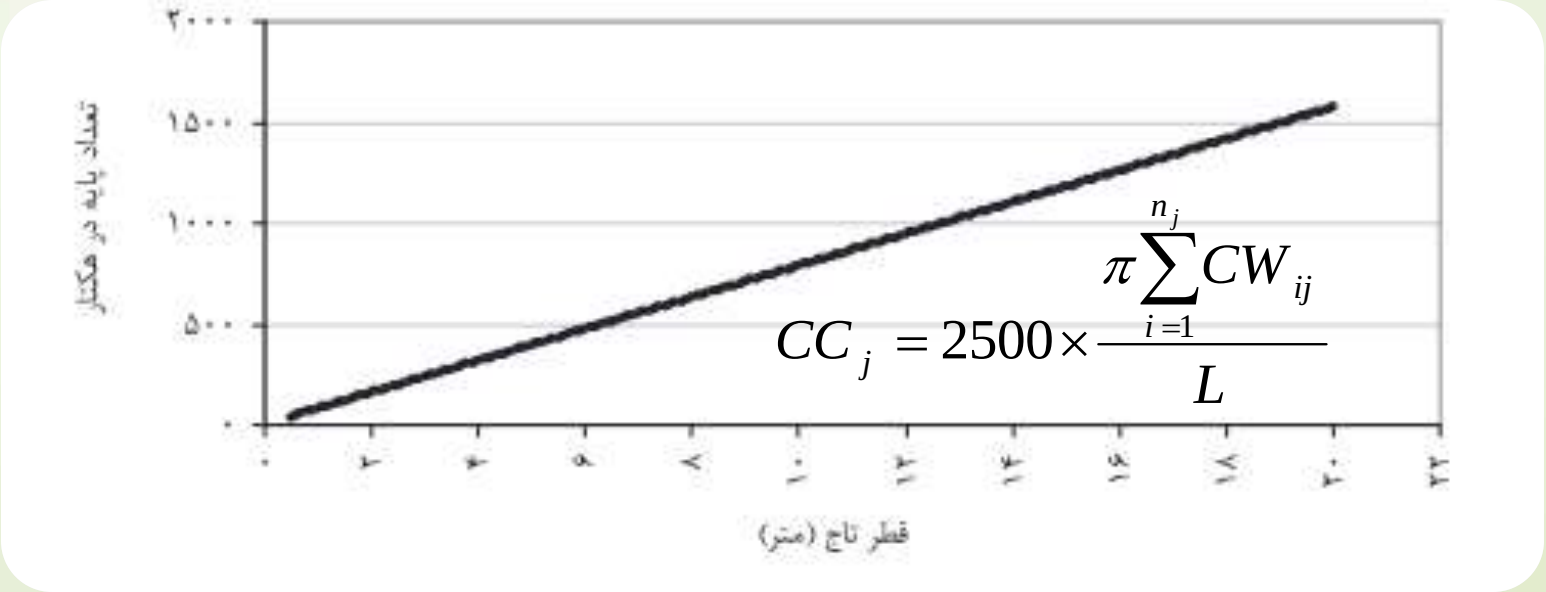
تحلیل اریبی



منشا ایجاد این اریبی ممکن است مرتبط با یك یا چند مورد از موارد زیر باشد:
اریبی ناشی از همپوشانی تاج درختان
اریبی ناشی از ساختار جنگل شامل درصد انبوهی و الگوی پراکنش مکانی درختان
اریبی ناشی از برآورد کننده

بحث اریبی ناشی از برآورد کننده





منابع این جلسه

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انتشارات دانشگاه تهران، چاپ دوم.
صفحات: 208-163

Ecological Methodology
***Krebs*, 1998**
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& Chapter 4: 139-141

Sampling Methods, Remote Sensing And GIS Multiresource Forest Inventory
***Kohl et al*. 2006**
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Chapter 3: 160-166

Optimization of variable area transect sampling using Monte Carlo simulation
Engeman and Sugihara
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A practical modification of horizontal line sampling for snag and cavity tree inventory
Ducey et al.
Canadian Journal of Forest Research 32: 1217–1224 (2002)

A new type of sample unit for the efficient assessment of diverse tree communities in complex forest landscapes
Sheil et al.
Journal of Tropical Forest Science 15 (1): 117-135 (2003)

Buffun’s needles problem

A thin needle of length *li* is placed on a flat area with a width of *W* and a length of *L* units, located within an area of irregular shape and size A. Parallel lines with a length of *L* units are drawn, one of them passing through the center.
It is assumed and required that the needle length *li* is less than the distance between two adjacent lines. The probability that the *j*th needle intersects the center line is a function of the needle length, the length of the sampling line, and the orientation of the needle. The latter is assumed to be a random variable, with a uniform distribution, within a domain extending between 0 and 180◦.



Georges-Louis Leclerc,
Comte de Buffon

The probability of intersection (de Vries 1986) is

$$p_i = \frac{2L \cdot l_i}{\pi A}$$

http://wiki.awf.forst.uni-goettingen.de/wiki/index.php/Line_sampling

Buffun’s needles problem

Introduction

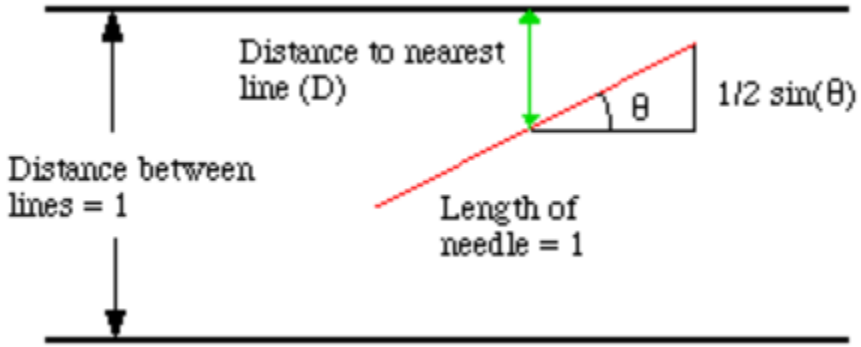
Buffon's Needle is one of the oldest problems in the field of geometrical probability. It was first stated in 1777. It involves dropping a needle on a lined sheet of paper and determining the probability of the needle crossing one of the lines on the page. The remarkable result is that the probability is directly related to the value of pi.

These pages will present an analytical solution to the problem along with a Java program for simulating the needle drop in the simplest case scenario in which the length of the needle is the same as the distance between the lines.

The Simplest Case

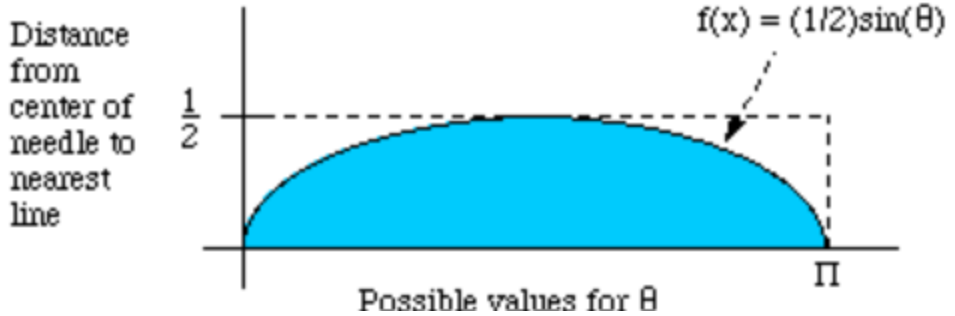
Let's take the simple case first. In this case, the length of the needle is one unit and the distance between the lines is also one unit. There are two variables, the angle at which the needle falls (theta) and the distance from the center of the needle to the closest line (D). Theta can vary from 0 to 180 degrees and is measured against a line parallel to the lines on the paper. The distance from the center to the closest line can never be more that half the distance between the lines. The graph below depicts this situation.

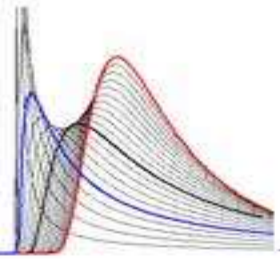
Buffun's needles problem



The needle in the picture misses the line. The needle will hit the line if the closest distance to a line (D) is less than or equal to 1/2 times the sine of theta. That is, D <= (1/2)sin(theta). How often will this occur?

In the graph below, we plot D along the ordinate and (1/2)sine(theta) along the abscissa. The values on or below the curve represent a hit (D <= (1/2)sin(theta)). Thus, the probability of a success it the ratio shaded area to the entire rectangle. What is this to value?



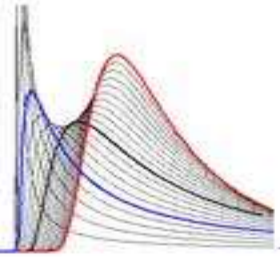


Buffun's needles problem

The shaded portion is found with using the definite integral of (1/2)sin(theta) evaluated from zero to pi. The result is that the shaded portion has a value of 1. The value of the entire rectangle is (1/2)(pi) or pi/2. So, the probability of a hit is 1/(pi/2) or 2/pi. That's approximately .6366197.
To calculate pi from the needle drops, simply take the number of drops and multiply it by two, then divide by the number of hits, or 2(total drops)/(number of hits) = pi (approximately)

The Other Cases

There are two other possibilities for the relationship between the length of the needles and the distance between the lines. A good discussion of these can be found in Schroeder .1974 ,The situation in which the distance between the lines is greater than the length of the needle is an extension of the above explanation and the probability of a hit is 2(L)/(K)pi where L is the length of the needle and K is the distance between the lines. The situation in which the needle is longer than the distance between the lines leads to a more complicated result.
http://mste.illinois.edu/reese/buffon/buffon.html



MVAT

If we examine the situation where stems are distributed randomly, uniformly and independently, the expected number of empty cells is described by an exponential relationship:

$$E(N_{empty}) = N_{total}\, e^{-Ax}$$

where

$E(N_{empty})$ is the mean number of empty cells per location,
N_{total} is the number of individual cells to be searched,
A is the per-cell search area (here the fraction of a hectare), and
x is the local stem density (n ha^{-1}).

For example given eight 10-m-wide cells to be searched over 15 m (i.e. N = 8, A =10 m X 15 m/10 000 m2 = 0.015 ha):

$$E(N_{empty}) = 8\, e^{-0.015x}$$

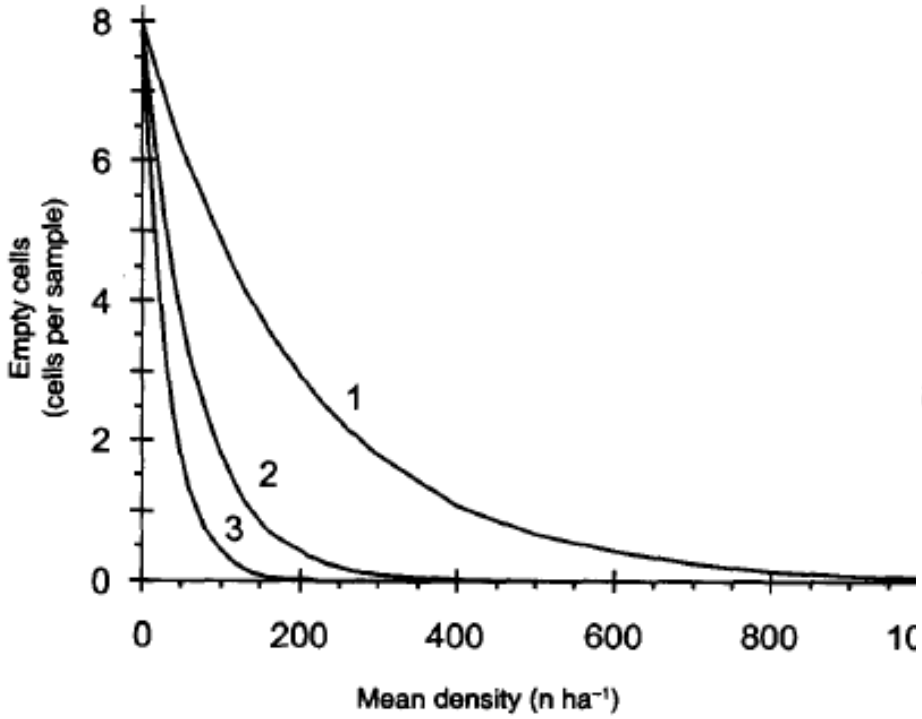
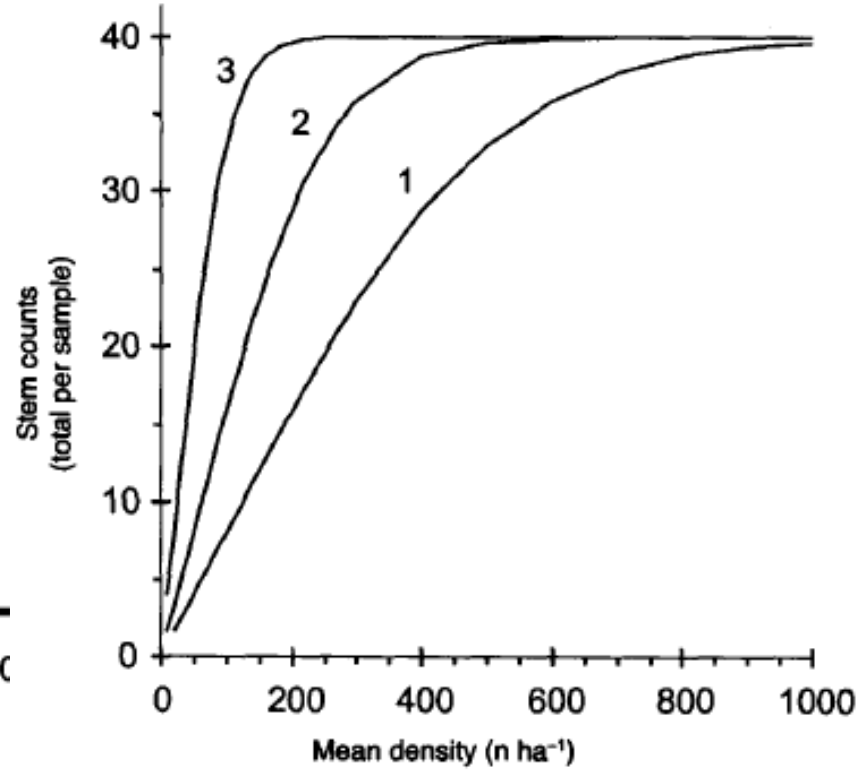
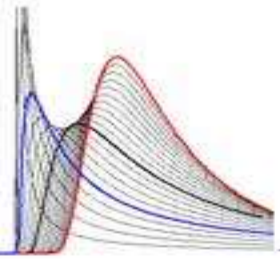


Figure 2 How the mean number of empty cells relates to stem-dens
when the members of the sample population are rand
distributed. Examples are given where the maximum se
distances are 5 m (1), 15 m (2) and 30 m (3).



re 3 How the mean number of stem-records relates to stem-densities when the members of the sample population are randomly distributed. Examples show results for 10-m-wide transects (cells) searched to a maximum of 10 m (1), 20 m (2) and 50 m (3).



MVAT estimator

For stand parameters to which individual trees contribute additively, such as basal area or biomass per hectare, we recommend the following procedure:

(1) Calculate the value of the variable of interest for each tree j in cell i. Call this value y_{ij}

(2) Multiply the y_{ij} values by their corresponding x_i values to obtain per unit area estimates for each tree. Sum these values of (x_i y_{ij}) over all trees in the cell to obtain the per unit area estimate for the cell. Call the estimate for cell i Y_i

(3) Average the Y_i values to obtain the best estimate for a group of cells.